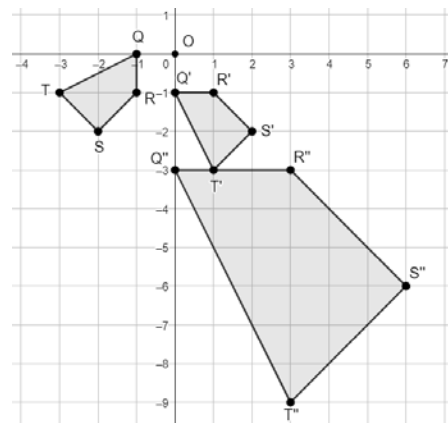


A sequence of transformations of $QRST$ is shown on the coordinate grid. $QRST$ is rotated then dilated, where $Q'R'S'T'$ is the rotation and $Q''R''S''T''$ is the dilation of $Q'R'S'T'$.



Complete the descriptions:

- $QRST$ was rotated 90° counterclockwise to create $Q'R'S'T'$.
- $Q'R'S'T'$ was dilated using a scale factor of 3 centered at (0,0) to create $Q''R''S''T''$.

3. Complete the table:

Transformation	Algebraic Description
Rotation	$(x, y) \rightarrow (-y, x)$
Dilation	$(x, y) \rightarrow (3x, 3y)$

4. Determine the coordinates if $QRST$ is first dilated then rotated:

Preimage $QRST$	
Q	$(-1, 0)$
R	$(-1, -1)$
S	$(-2, -2)$
T	$(-3, -1)$

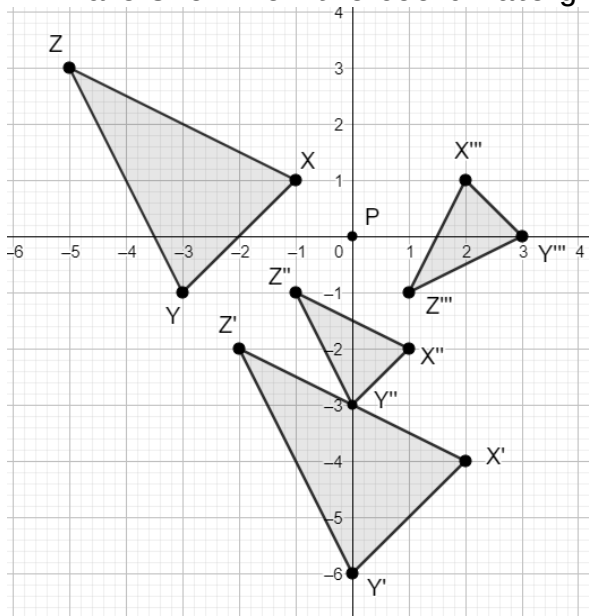
Dilation $Q'R'S'T'$	
Q'	$(-3, 0)$
R'	$(-3, -3)$
S'	$(-6, -6)$
T'	$(-9, -3)$

Rotation $Q''R''S''T''$	
Q''	$(0, -3)$
R''	$(3, -3)$
S''	$(6, -6)$
T''	$(3, -9)$

- Based on the results above, the sequence of transformations commutative.

☒ is
☐ is not

$XYZ, X'Y'Z', X''Y''Z'',$ and $X'''Y'''Z'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- XYZ is first

☒ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'Y'Z'$, then

☐ translated
☐ reflected
☐ rotated
☒ dilated

 to $X''Y''Z''$,
- and finally

☐ translated
☐ reflected
☒ rotated
☐ dilated

 to $X'''Y'''Z'''$.
7. Describe the transformation for $XYZ \rightarrow X'Y'Z'$.
Translate 5 units down, 3 units right
8. Describe the transformation for $X'Y'Z' \rightarrow X''Y''Z''$.
Dilate with a scale factor of $\frac{1}{2}$
9. Describe the transformation for $X''Y''Z'' \rightarrow X'''Y'''Z'''$.
Rotate 90° counterclockwise

10. Write the algebraic description for the sequence of transformations from 7-9:

Transformation	Algebraic Description
Translation	$(x,y) \rightarrow (x+3,y-5)$
Dilation	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$
Rotation	$(x,y) \rightarrow (-y,x)$